UNIVERSITY OF CAPE COAST
DEPARTMENT OF MATHEMATICS
COURSE OUTLINE

MAT 101: Algebra and Trigonometry **Semester: 1, 2026/27** **3 Credits**

Lecturer: Dr. (Mrs.) Henrietta Nkansah
Dr. Albert Lanor Sackitey
Dr. Nicholas Opoku

Office Hours: Monday to Friday,
9:00am – 2:00pm

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Prerequisite Course(s): SSSCE Elective Mathematics or WASSCE Elective Mathematics
Recommended Text(s):

1. Bird J. (2010). *Higher Engineering Mathematics*, 6th Edition. Oxford: Newnes.
2. Backhouse, J. K. and Houldsworth, S. P. T. (2005). *Pure Mathematics 1*. Longman. UK.
3. Essel, E. K. and Nkansah, B. K. (2013). *Algebra and Trigonometry*. University Press, Cape Coast.
4. Larson, R., (2011). *Algebra and Trigonometry*, Eighth Edition, Brooks/Cole Publishing Co., Belmont, USA.
5. Smith, K. L. (1988). *College Mathematics and Calculus with Applications to Management, Life and Social Sciences*; Brooks/Cole Publishing Co. California. USA.
6. Sullivan, M. (2012). *Algebra & Trigonometry*. 9th ed. Pearson Education, Inc. Boston

Course Description

The course gives basic knowledge in algebra and trigonometry. This knowledge prepares students for further studies in Mathematics and other specialized courses in Mathematical sciences and other disciplines such as Physics, Biology, Economics and Business. It also provides solid foundation for critical thinking when solving abstract proves for the validation of Mathematical beliefs.

Course Objectives

At the end of this course, students will be able to:

- identify and use the properties of union and intersection of sets;
- perform basic operations on sets;
- carry out operations on real numbers;
- determine various features of polynomial functions;
- perform operations on rational functions;
- identify and draw the graphs of functions;
- apply the rules of indices and logarithms to solve problems;
- formulate and solve linear programming problems;
- write down the nth term of a sequence and the sum of the first n term of any given series;
- use various trigonometric identities to solve simple trigonometric equations.

Course Content

Unit 1: Sets
Unit 2: The Real Number System
Unit 3: Linear and Quadratic Functions
Unit 4: Polynomial and Rational Functions
Unit 5: Linear Inequalities and Programming
Unit 6: Binomial Theorem
Unit 7: Logarithmic Functions
Unit 8: Series and Sequences
Unit 9: Trigonometric Functions

Teaching and Learning Strategies

- Lectures
- Discussions
- Identification and Solving of Problems
- Group work
- Question and Answers

Assessment

Assessment including quizzes, assignments and examination will be used.

Assessment weighting

Quiz 1	15%
Quiz 2	15%
Assignment (Individual or Group)	10%
End-of-semester Examination	60%

You will be graded as follows:
A=80-100%; **B+**=75-79%; **B** =70-74%, C+ =65-69%, **C**= 60-64%, D+ =55-59%, D= 50-54%, **E< 50** (Fail)

Quiz 1: 25th October, 2026 Weeks 1-4
Quiz 2: 29th November, 2026 Weeks 5-8
Final examination covers the entire syllabus.

Course Outline

Week	***Content***	***Objectives*** *Students will be able to:*	***Activities***
1	Discussion of course outline	outline the areas to be covered in the course and the criteria for assessment, also explain the course policy and what is generally expected of them in the course	Discussion
2-3	Sets	- Define and give examples of Commutative, associative and distributive properties of union and intersection of sets. - De Morgan's laws of union and intersection of sets. - Determine and sketch the Cartesian products of sets	Discussion/ Lecture

4-5	The real number system	natural numbers, integers, rational and irrational numbers. Properties of addition and multiplication on the set of real numbers. Relation of order in the system of real numbers	Lecture
6-7	Polynomial and Rational Functions	Classify functions as linear, quadratic and other polynomial functions, rational algebraic functions, absolute value functions, functions containing radicals and their graphical representation.	Lecture/ Discussion
8	Linear Programming	Inequalities in one and two variables. Application to linear programming Binomial theorem for integral and rational indices	Lecture/ Group Assignment Project Work
9	Logarithmic Functions Binomial Theorem	Indices and logarithm, their laws and applications. Binomial theorem for integral and rational indices	Discussions Solving Problems
10	Series and Sequences	Arithmetic Progression Geometric Progression Mixed Sequences Recursive	Lecture Discussion Problem Solving
11-12	Trigonometric Functions	Circular functions of angles of any magnitude and their graphs. Prove trigonometric formulae including identities. Solve trigonometric equations applying the concepts of compound, multiple angles, half angles and triple angles	Lecture Discussion Solve problems involving Trig. Demonstrate the applications of Trigonometry to real life problems
12+	**REVISION**		
	END OF SEMESTER EXAMS		